

• GEOMETRIC RANDOM VARIABLES

prob. of success on kth trial and not before.

Let N = no. of coin tosses until 1st head

$P_N(n) = (1-p)^{n-1} \cdot p$ PMF

e.g. $N=11 \Rightarrow (1-p)^{10} p$

CDF: $F_N(x) = \sum_{n=1}^x P_N(n)$

• TAIL CDF = 1 - CDF

$\bar{F}_N(k) = 1 - F_N(k) = P(N > k)$

$$\begin{aligned} P(N > k) &= \sum_{n \geq k+1} P(N = n) \\ &= (1-p)^k p [1 + (1-p) + (1-p)^2 + \dots] \\ &= (1-p)^k \end{aligned}$$

$$P(n > k+m \mid n > k) = \frac{(1-p)^{k+m}}{(1-p)^k} = (1-p)^m$$

independent of k $P(A|B) = \frac{P(A \cap B)}{P(B)}$

$= P(N > m) !!!$
MEMORYLESS PROPERTY

• POISSON RANDOM VARIABLES

= comes with a parameter λ

PMF: $p_X(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$ mean = λ

• Induced measure P_X and CDF

$$P(x \leq x_1) = P_X(\underbrace{(-\infty, x_1]}_{\text{subset of } \mathcal{B}(\mathbb{R})}) = \mathbb{P}\{\omega \in \Omega : X(\omega) \leq x_1\}$$

$\therefore \text{CDF} = F_X(x_1) = P_X((-\infty, x_1])$

If $F_X(\cdot)$ is continuous, then X is continuous.
If $F_X(\cdot)$ is piecewise constant, then X is discrete.

$$\left[\begin{array}{l} F_X(\infty) = 1 \\ F_X(-\infty) = 0 \end{array} \right] \rightarrow \text{CONDITION: } X \in (-\infty, \infty)$$

* $F_X: \mathbb{R} \rightarrow [0, 1]$ is non-decreasing and right continuous.
At a point of discontinuity,
 $F_X(x) = F_X(x+)$

• CONTINUOUS RANDOM VARIABLES

$F_X(\cdot)$ is continuous $\Rightarrow X$ is a continuous RV

Probability density function (pdf) = PROBABILITY PER UNIT LENGTH

$$f_X(x) = \lim_{\Delta \rightarrow 0^+} \frac{P(x < X < x + \Delta)}{\Delta} = \lim_{\Delta \rightarrow 0^+} \frac{F_X(x + \Delta) - F_X(x)}{\Delta} = \frac{dF_X(x)}{dx}$$

$$\underbrace{F_X(x)}_{\text{cdf}} = \int_{u=-\infty}^x \underbrace{f_X(u)}_{\text{pdf}} \cdot du$$

* $P_X(\mathbb{R}) = \int_{u=-\infty}^{\infty} f_X(u) \cdot du = 1$

* $P_X(a \leq X \leq b) = \int_a^b f_X(u) \cdot du$

* In general, $P_X(B) = \int_{u \in B} f_X(u) \cdot du$

$$E[X] = \int_{-\infty}^{\infty} u \cdot f_X(u) \cdot du$$

$$E[X^n] = \int_{-\infty}^{\infty} u^n \cdot f_X(u) \cdot du$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(u) \cdot f_X(u) \cdot du$$

* probability that a continuous r.v takes a particular value = 0
 $\therefore$ we integrate the PDF over an interval.